

Mortality Equations for CONUS

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Data

We took the FIA mortality data subset, filtered out missing data (primarily crown ratio and heights). This left 4343197 observations. We found a number of observations with large decreases in crown closure at tree tip (CCH). For this analysis, we set the change in CCH to 0.0 for those observations by making the ending measurement equal to the initial measurement.

The observations are distributed among states is show in Table 1.

Table 1: Mortality Observations by State

State	n
MN	396255
ME	327368
WI	321417
MI	298910
OR	204776
GA	176540
NY	170537
PA	152055
AL	141930
WA	137839
VA	136219
MO	135853
NC	124516
AR	112689
SC	112014
MS	111427
TN	109749
CA	105913
MT	95025

Table 1: Mortality Observations by State

State	n
KY	88127
TX	85291
CO	79722
WV	78684
NH	73331
ID	61960
OH	53553
VT	52214
FL	49165
LA	43621
IN	42858
OK	36910
IL	30733
MA	24706
UT	24542
IA	17511
AZ	16740
NM	15355
SD	13697
NJ	13336
MD	13040
WY	11708
CT	10842
KS	10777
NE	7360
ND	5496
RI	4712
DE	4444
NV	1730

Base Rate Mortality

Daniel Kanneman and Amos Tversky¹ made the point that any putative model should improve upon the *base rate*. In our case, the base rate is the average mortality rate for each species accounting for no other factors. We calculate this by taking the annualized mean mortality rate for each species in the data set.

¹Kahneman D. and Tversky, A. (1973). On the Psychology of Prediction. *Psychology Review*. 80(4), 237-251. doi: 10.1037/h0034747

The base rate estimates for 183 species are distributed as shown in Figure 1. Most base rate annual survival estimates are between 98 and 100%.

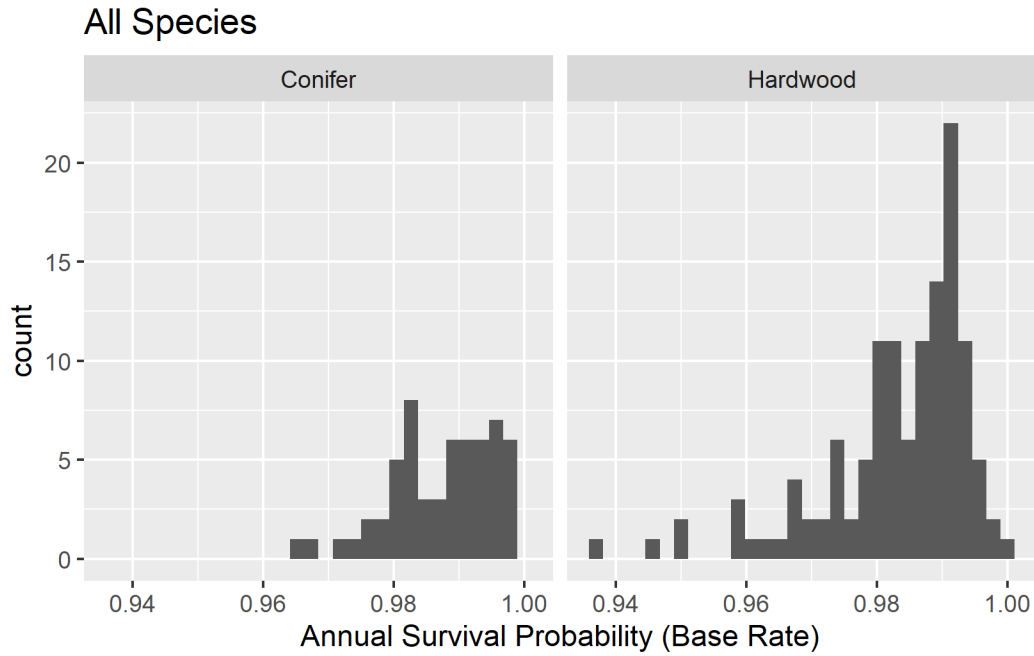


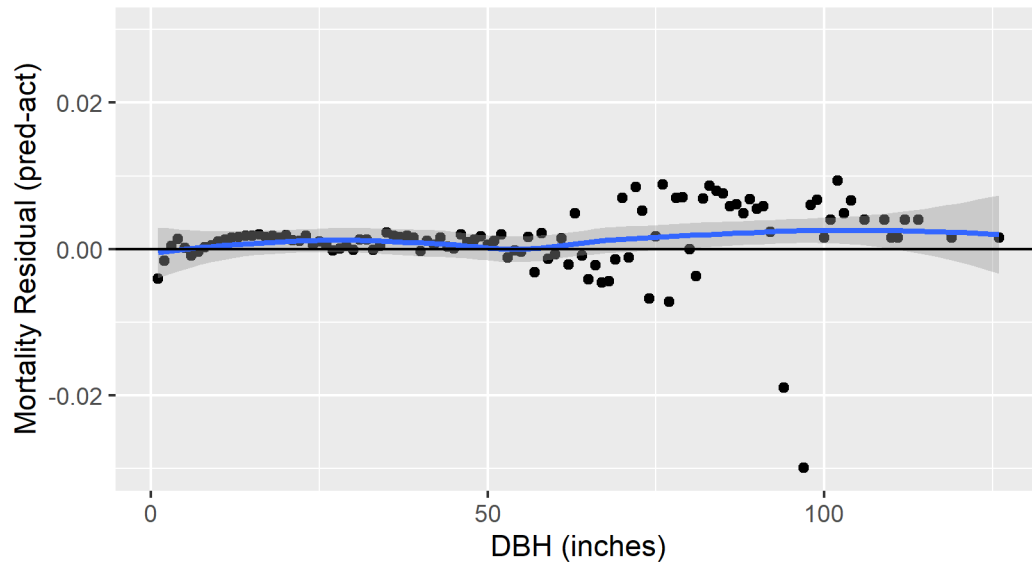
Figure 1

Base Rate Residuals

The following residual graphs show the average errors for all species over classes of possible explanatory variables.

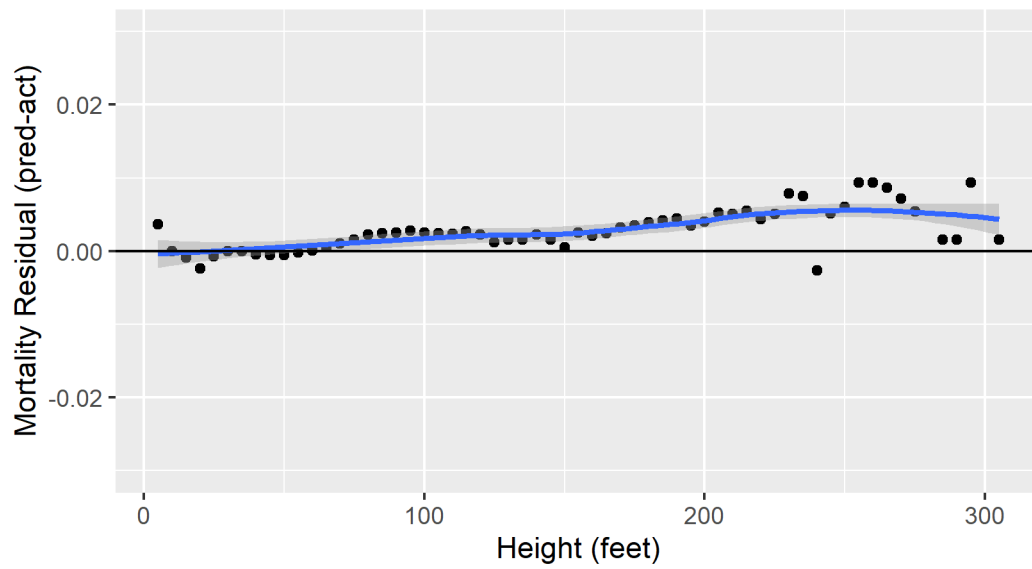
Mortality Residuals

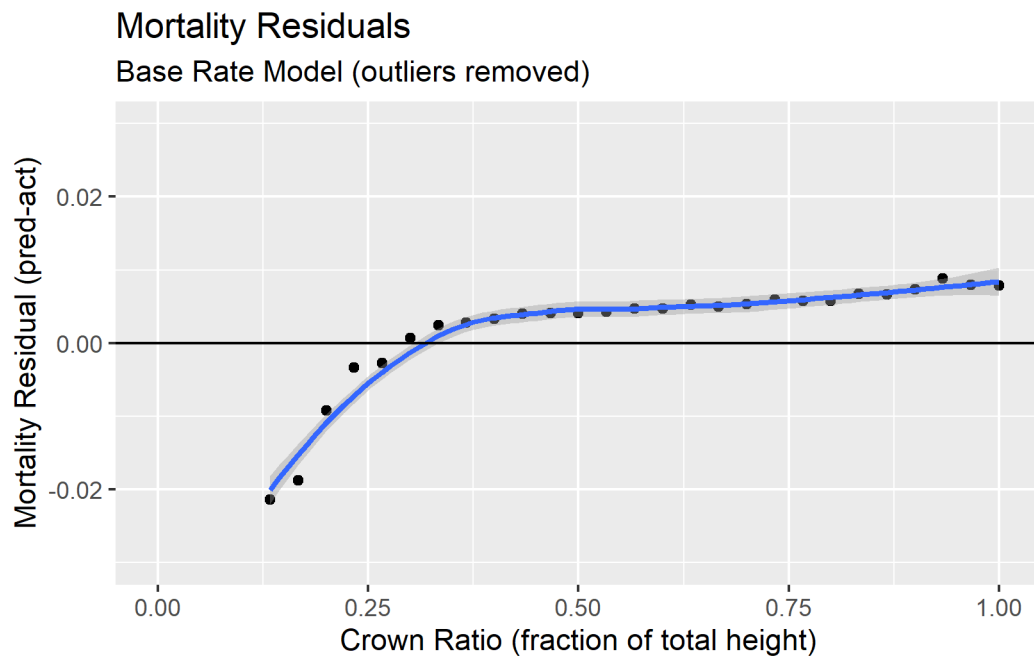
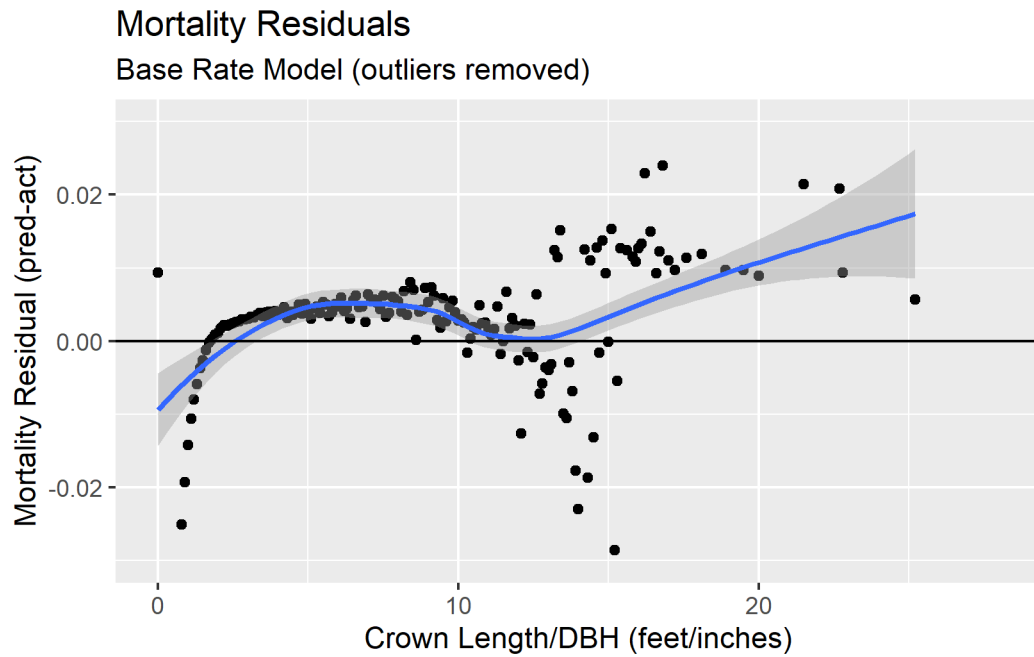
Base Rate Model (outliers removed)

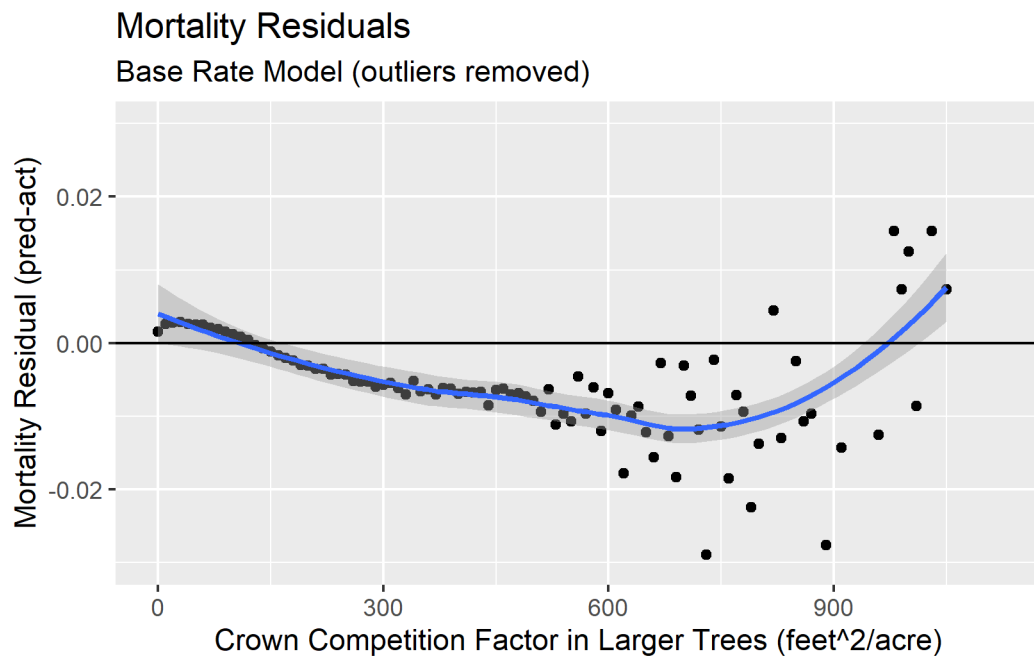
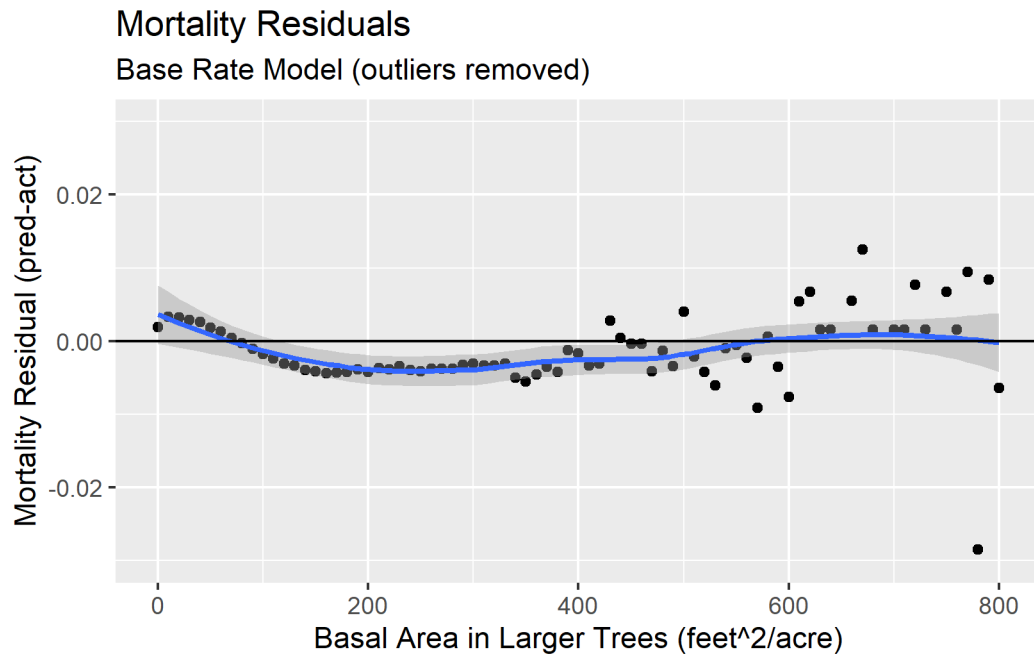


Mortality Residuals

Base Rate Model (outliers removed)

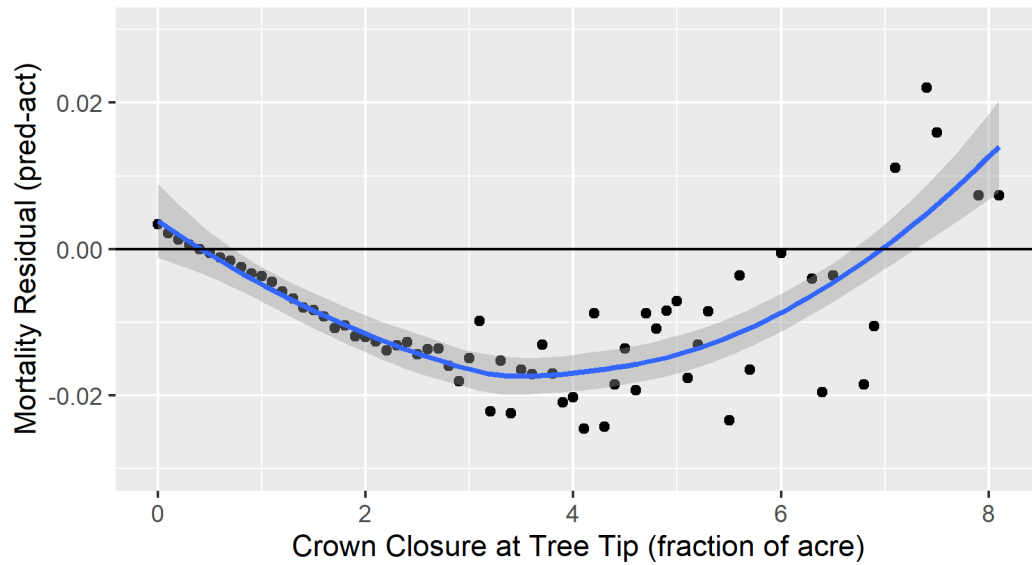






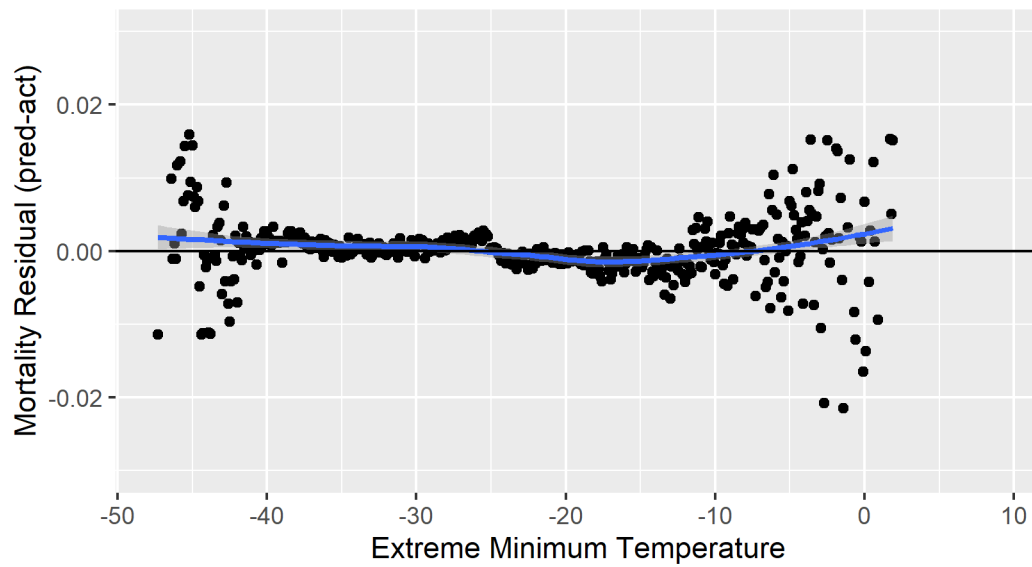
Mortality Residuals

Base Rate Model (outliers removed)



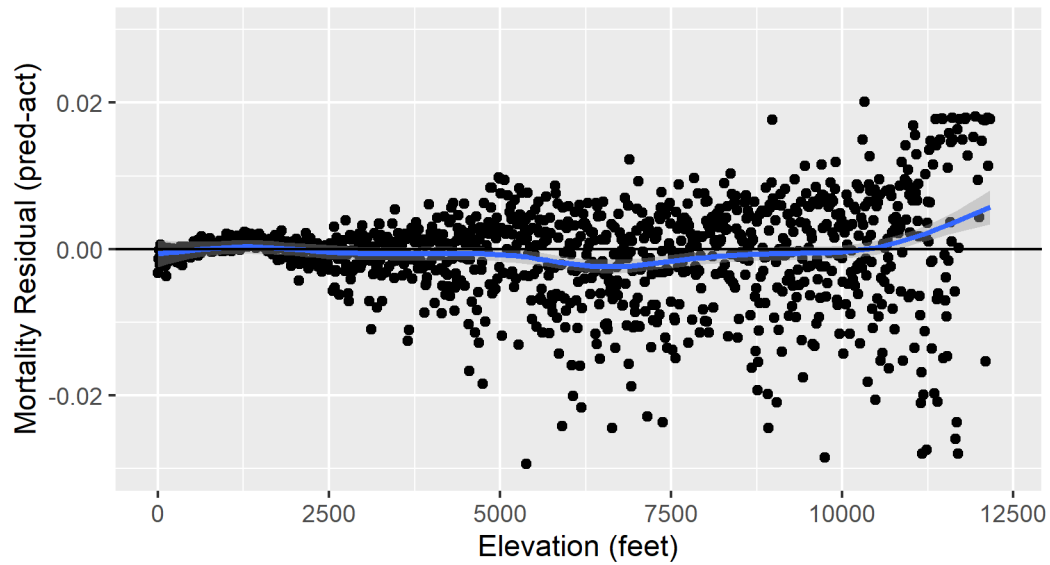
Mortality Residuals

Base Rate Model (outliers removed)



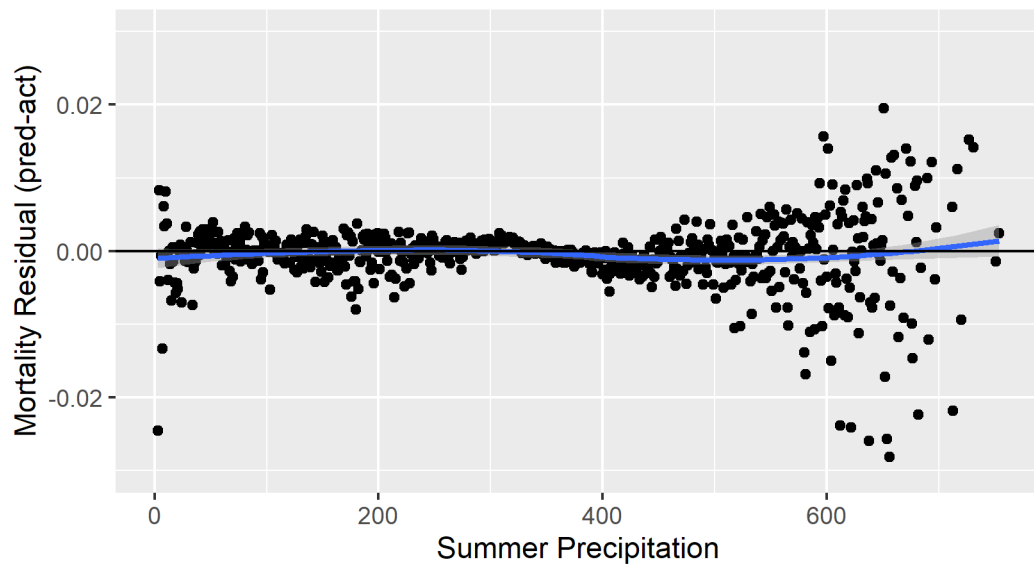
Mortality Residuals

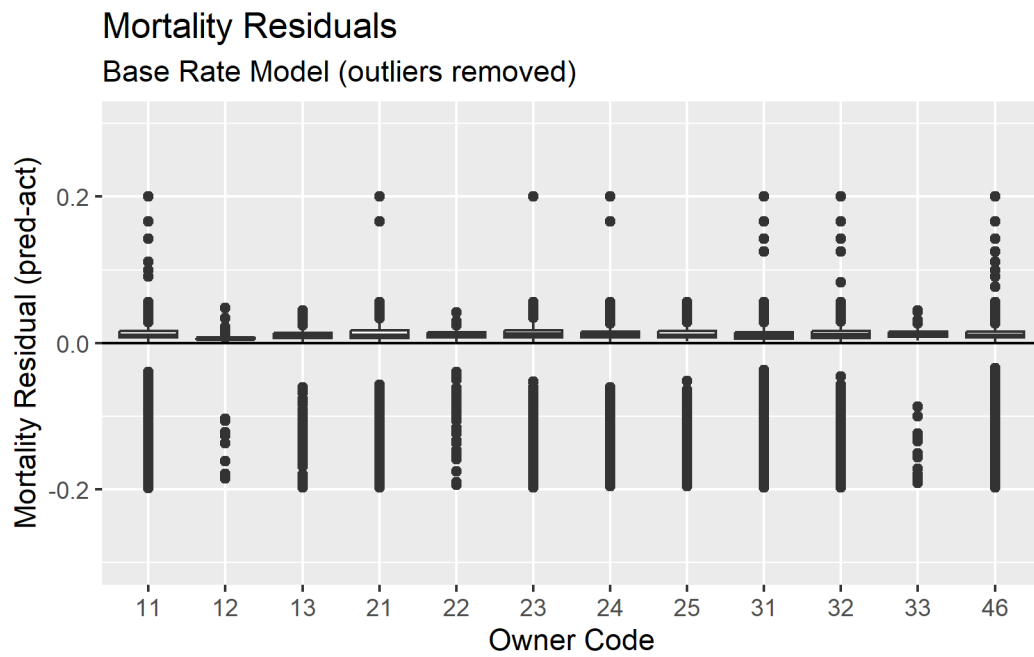
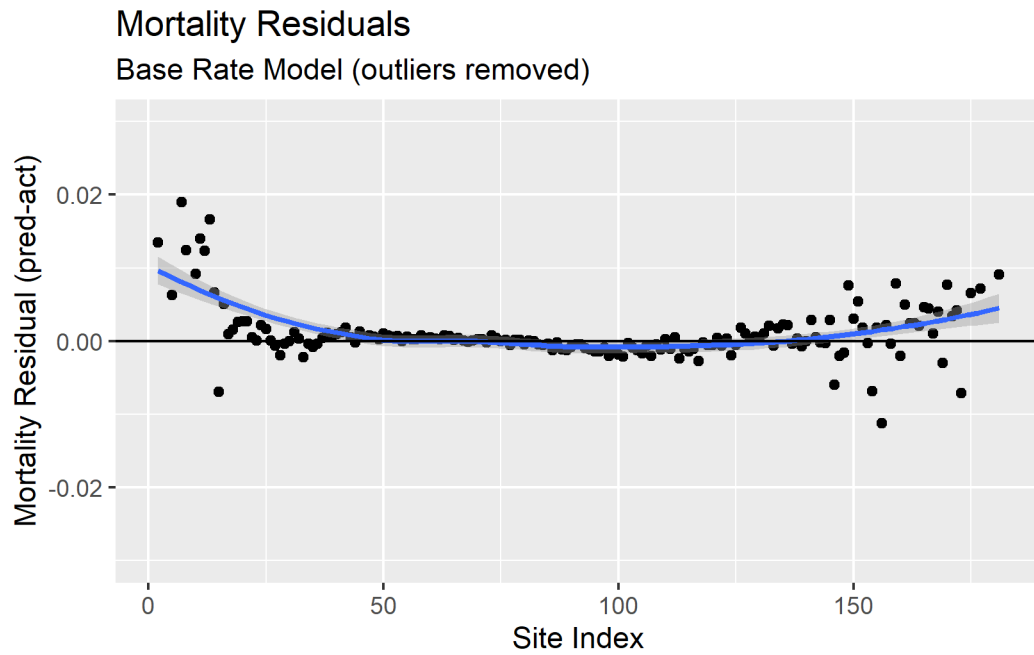
Base Rate Model (outliers removed)

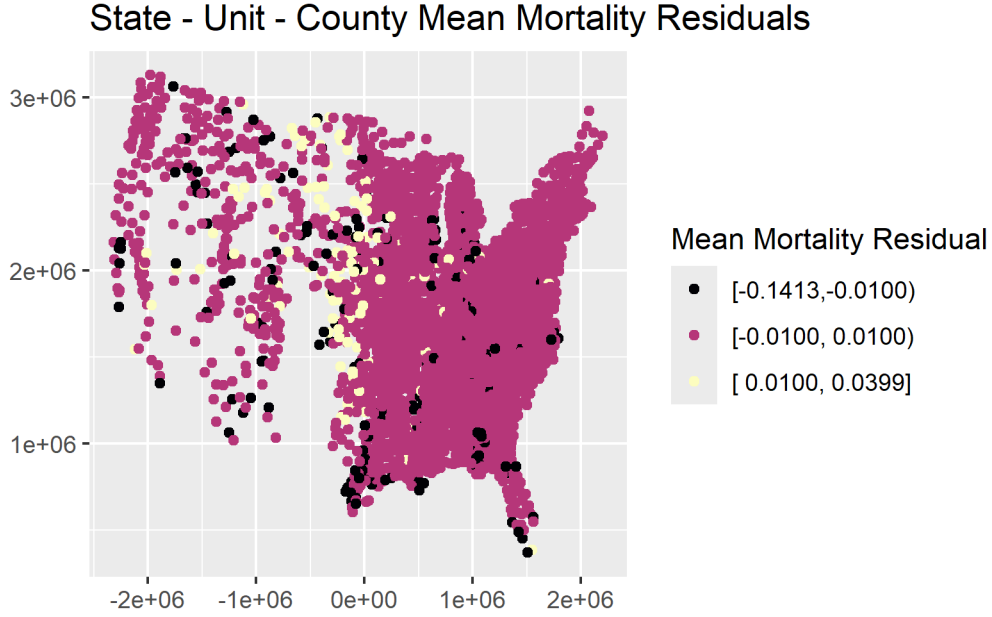


Mortality Residuals

Base Rate Model (outliers removed)







Proposed Model

The model we propose to explain more of the mortality variation over the base rate is a Gompit function:

$$P_{surv} = 1 - e^{(-e^{(\beta_0 + \beta_1(cr + 0.01)^{\beta_2} + \beta_3 cch^{\beta_4}))})} \quad (1)$$

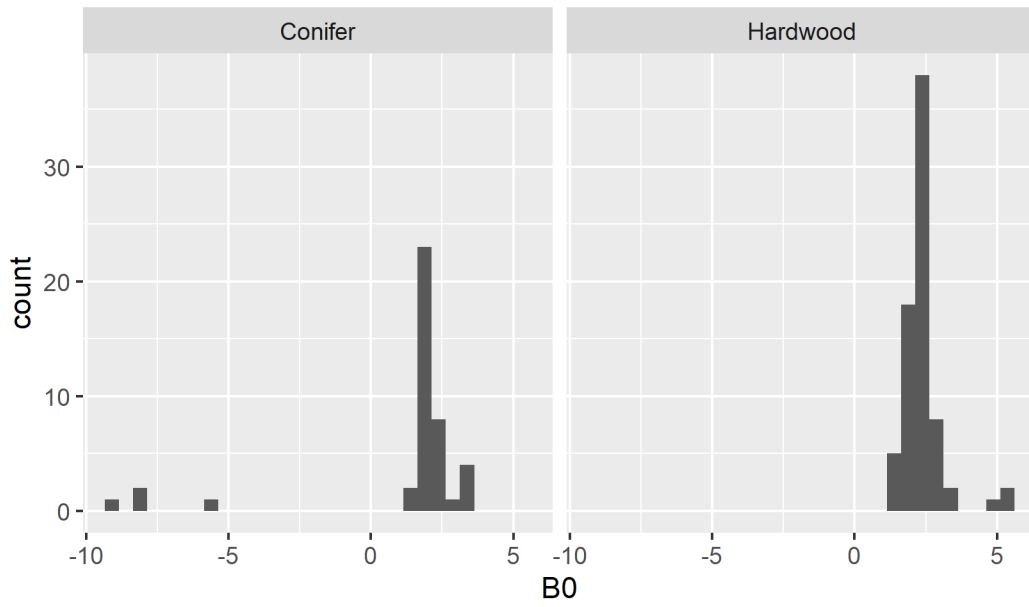
where:

- P_{surv} : probability of survival
- cr : crown ratio
- cch : crown closure at tree tip

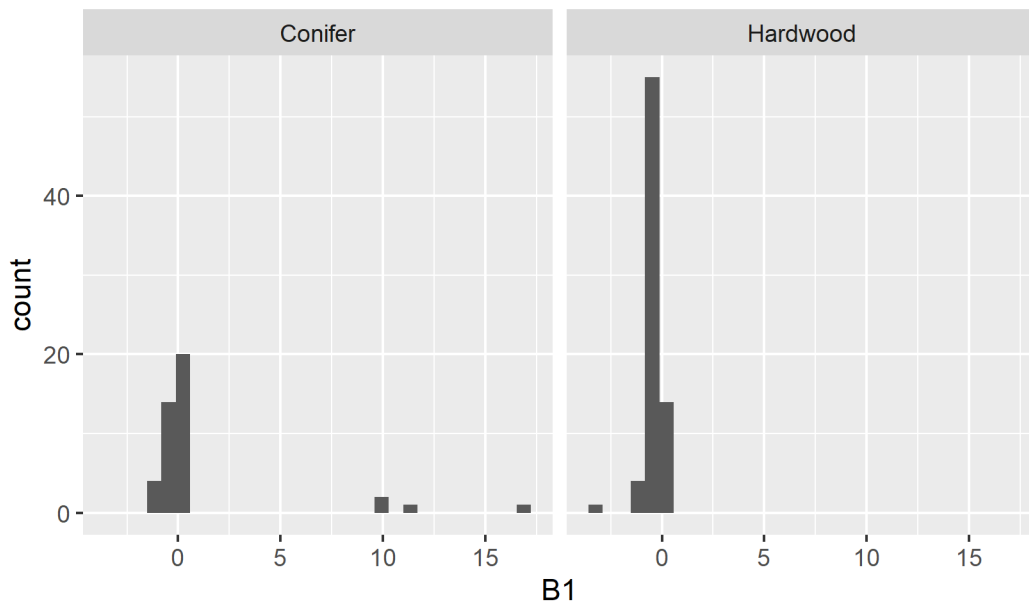
We used an integrative fitting approach to account for varying period lengths and we minimized the negative log-likelihood for each species separately.

We were able to make parameter estimates for 116 species. The distributions of the estimates for these species are shown below.

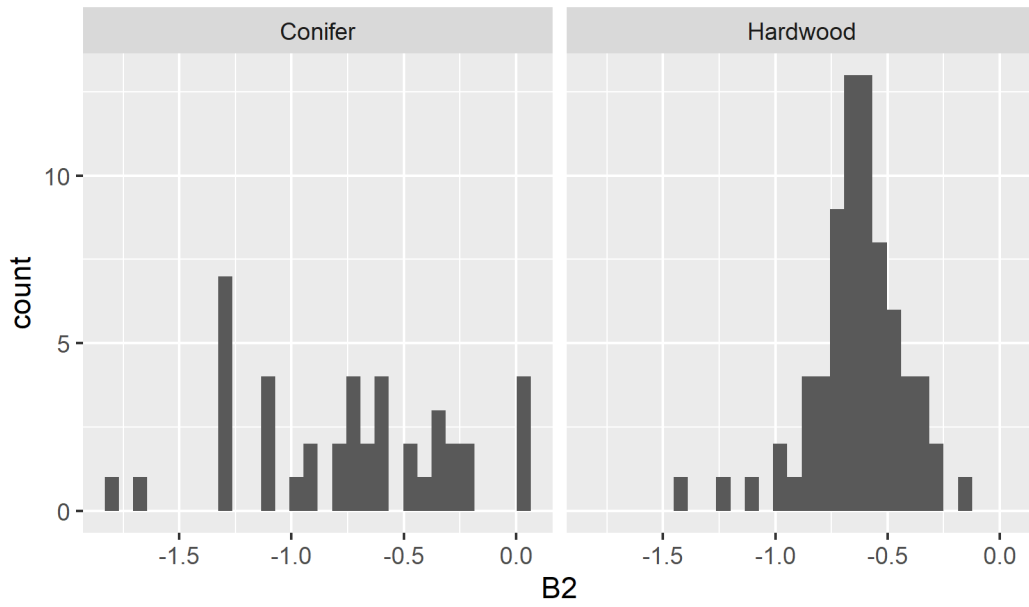
All Species



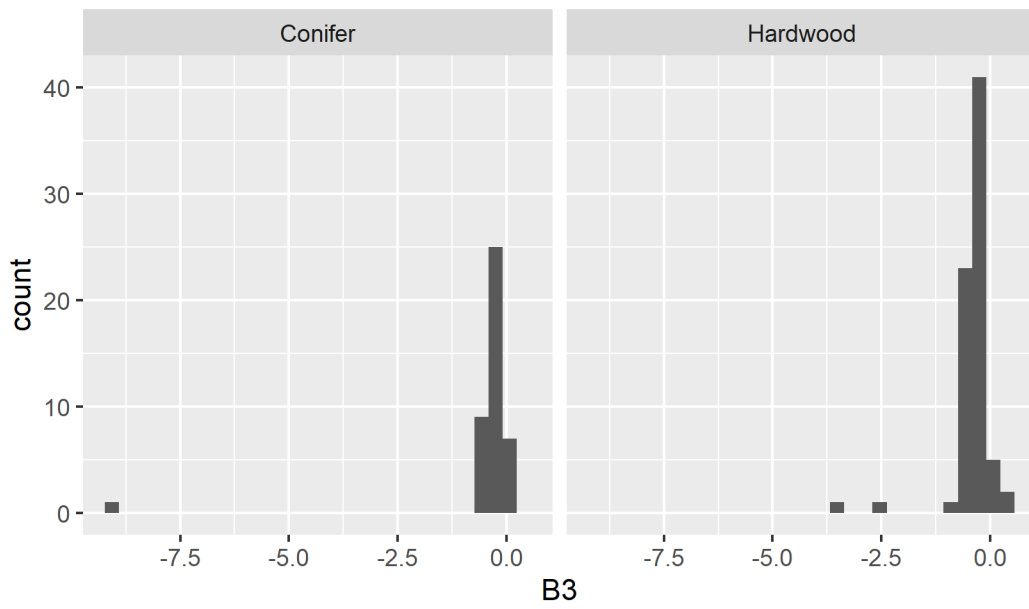
All Species



All Species



All Species



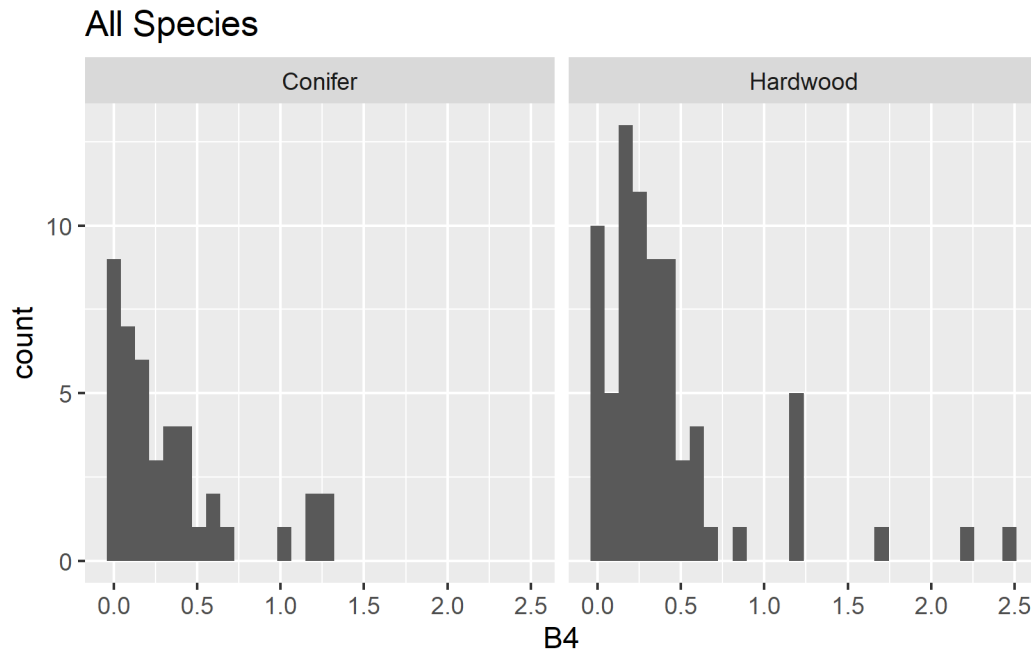


Figure 2 shows that all species' negative log likelihoods were reduced with the proposed model.

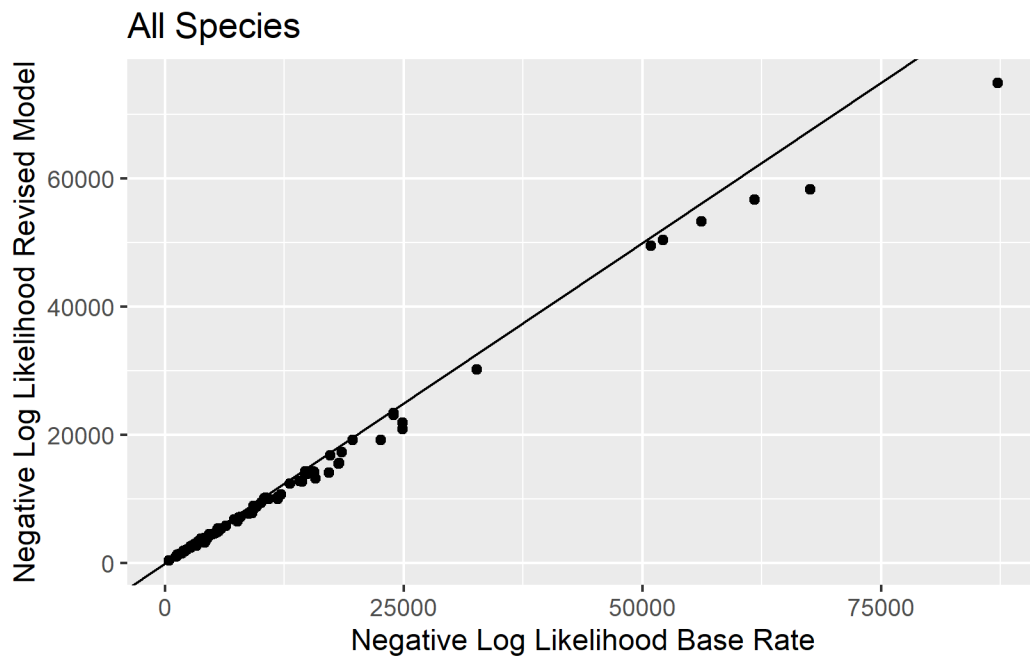


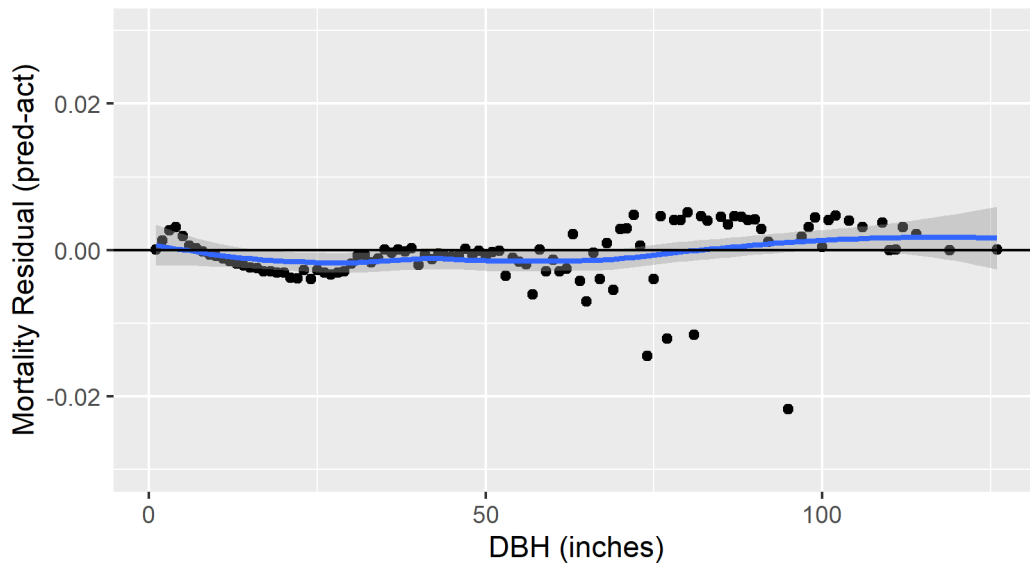
Figure 2

Proposed Model Residuals

The following residual graphs show the average errors for all species where the fit improved over the base rate over classes of possible explanatory variables.

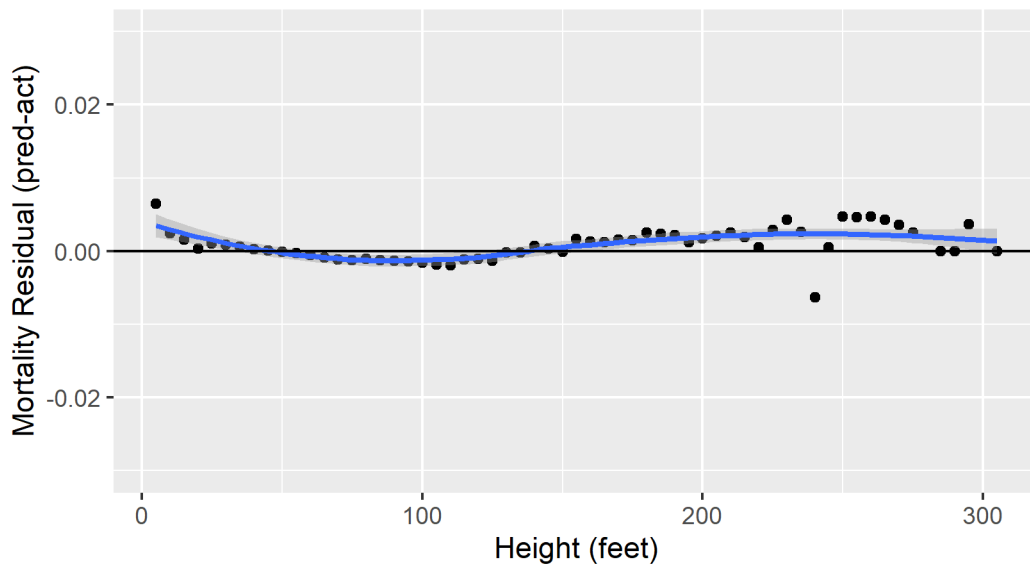
Mortality Residuals

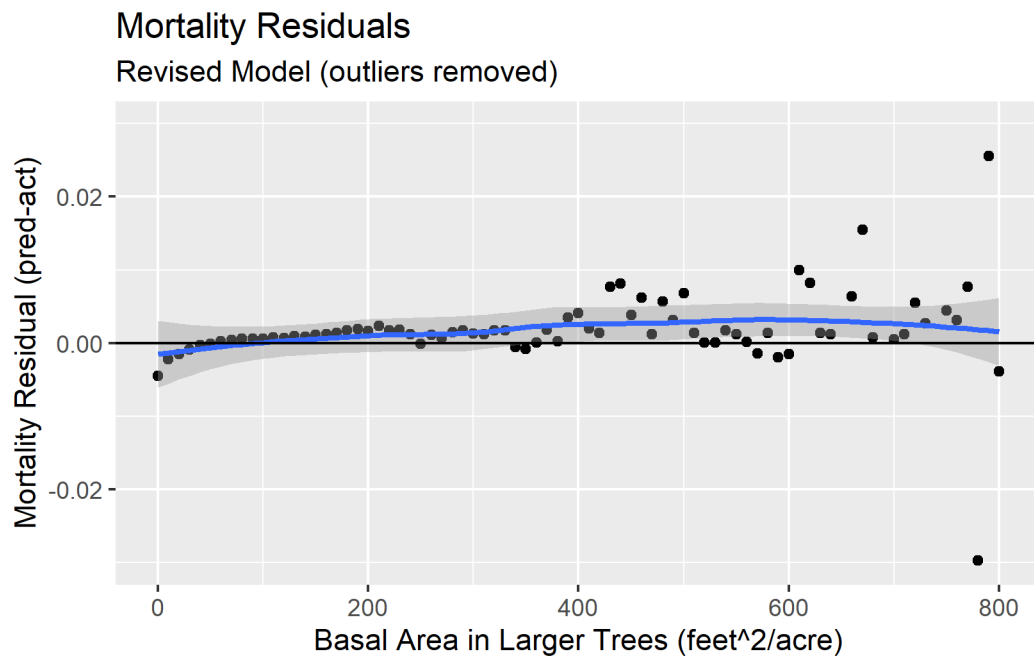
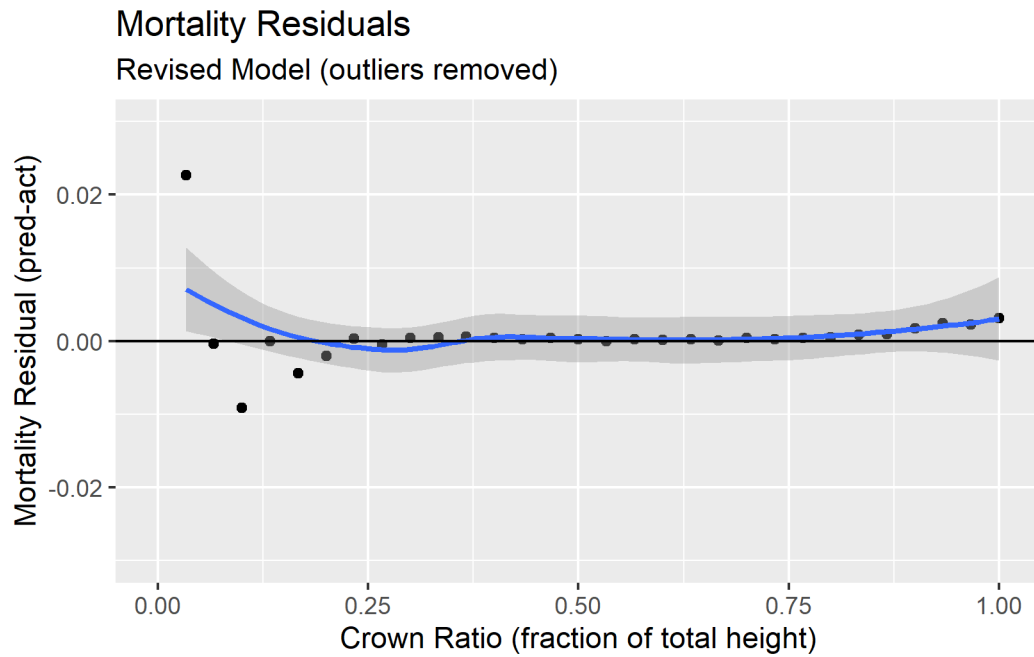
Revised Model (outliers removed)

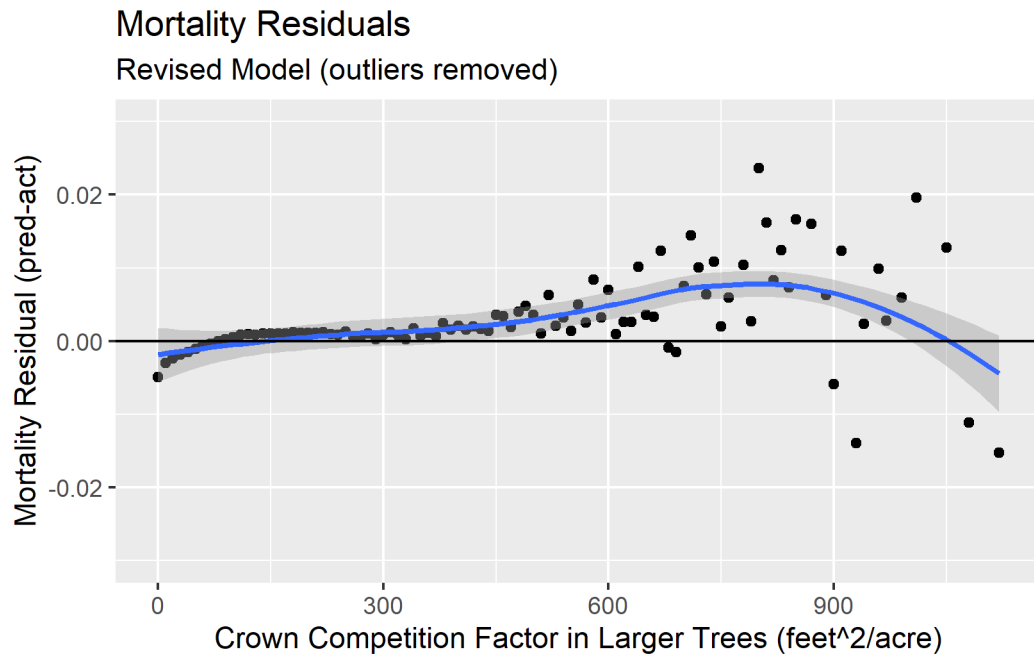


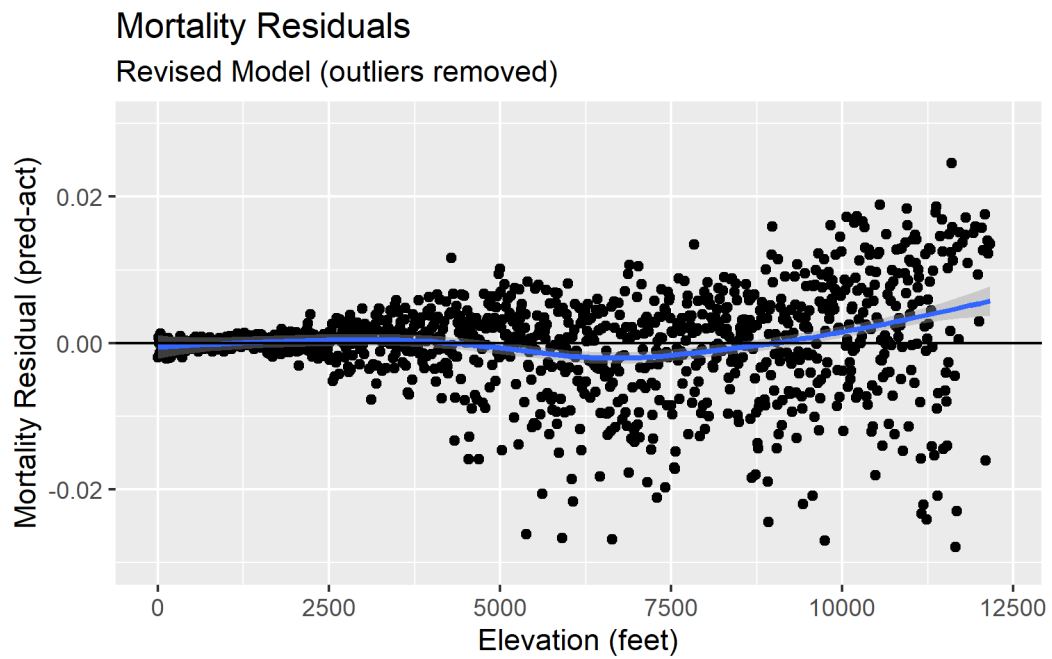
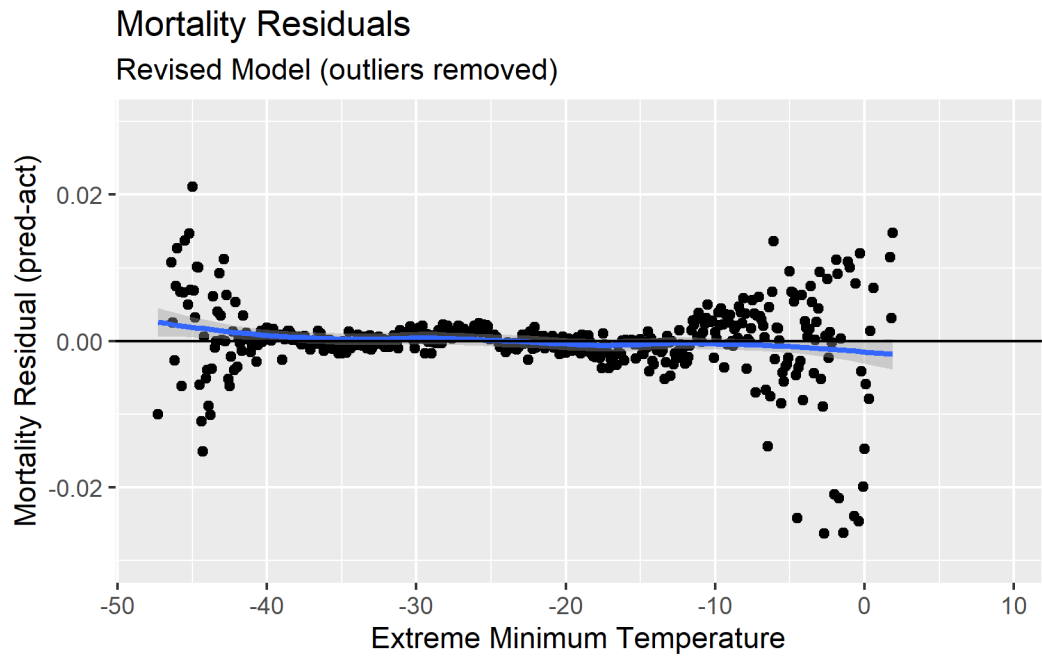
Mortality Residuals

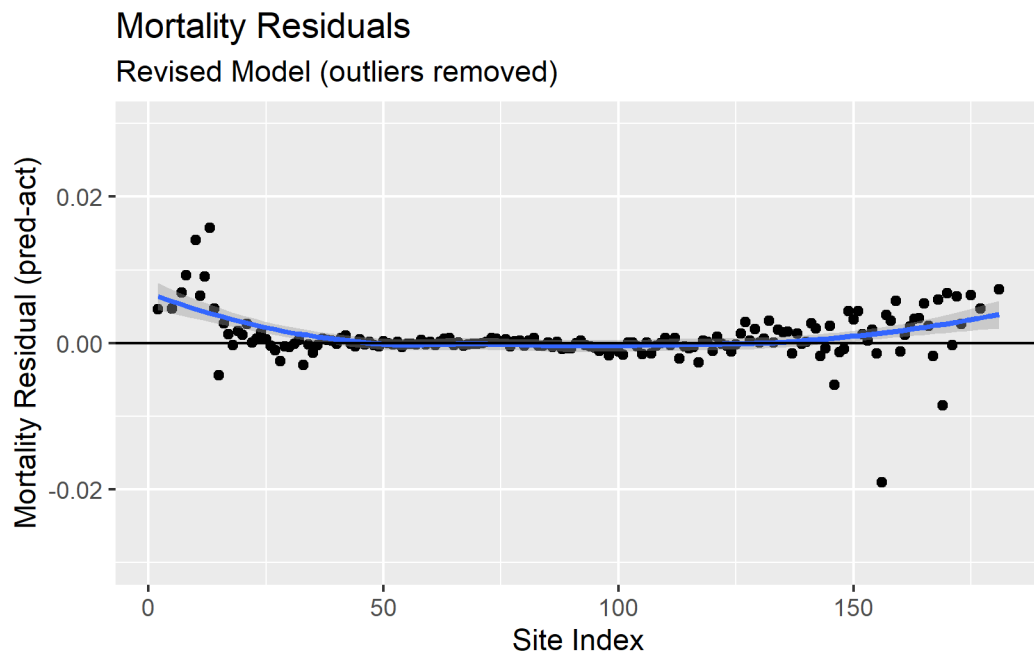
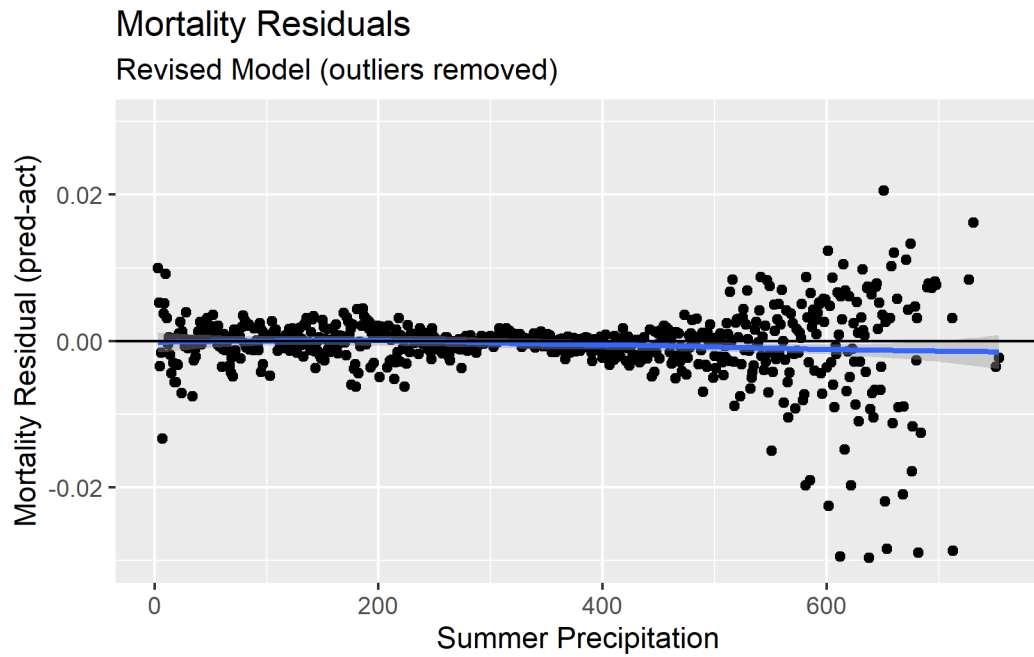
Revised Model (outliers removed)



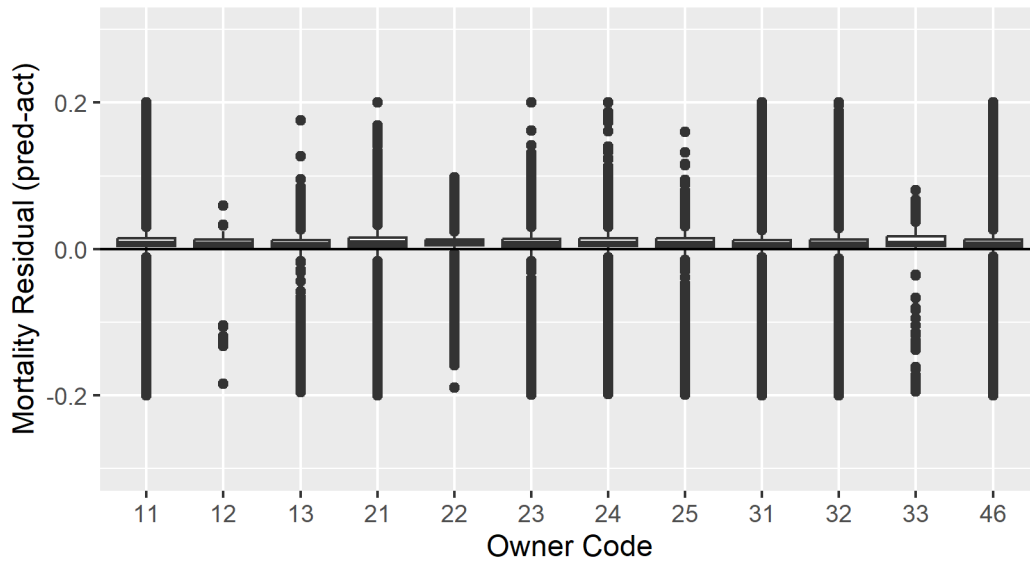








Mortality Residuals
Revised Model (outliers removed)



State - Unit - County Mean Mortality Residuals

